

Molecule Builder

Level 1: Starter Molecules

Build each one, then check the box. Count your parts first!

○ Hydrogen (H) ● Carbon (C) ● Oxygen (O) ● Nitrogen (N) ● Chlorine (Cl) — single bond — double/triple = bendy bonds

1 Hydrogen gas

H₂

☐ Built it!



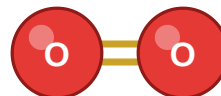
○ 2 Hydrogen — 1 straight

The lightest stuff in the universe. The Sun is mostly hydrogen!

2 Oxygen gas

O₂

☐ Built it!



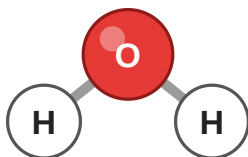
● 2 Oxygen — 0 straight — 2 bendy

You breathe this in with every breath.

3 Water

H₂O

☐ Built it!



Shape: Bent (V shape)

● 1 Oxygen ○ 2 Hydrogen — 2 straight

Bent like a boomerang. Every drop has trillions of these!

4 Carbon dioxide

CO₂

☐ Built it!



Shape: Straight line

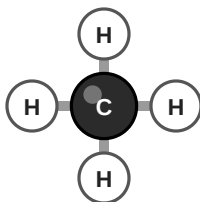
● 1 Carbon ● 2 Oxygen — 0 straight — 4 bendy

You breathe it out, plants breathe it in. It makes soda fizzy!

5 Methane

CH₄

☐ Built it!



Shape: Pyramid (tetrahedron)

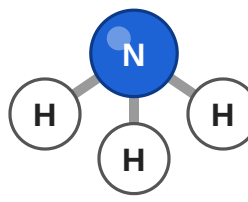
● 1 Carbon ○ 4 Hydrogen — 4 straight

Natural gas for the stove. Cows burp lots of it!

6 Ammonia

NH₃

☐ Built it!



Shape: Short pyramid

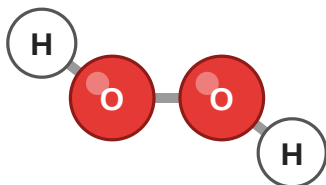
● 1 Nitrogen ○ 3 Hydrogen — 3 straight

Smelly cleaner. Plants need nitrogen like this to grow.

7 Hydrogen peroxide

H₂O₂

☐ Built it!



● 2 Oxygen ○ 2 Hydrogen — 3 straight

The stuff that fizzes when you clean a scraped knee.

8 Hydrogen chloride

HCl

☐ Built it!



● 1 Chlorine ○ 1 Hydrogen — 1 straight

Mixed with water it becomes the acid in your stomach that digests food.

Tip: kit colors can vary. Match the colors to the key card in your kit. Double bonds use 2 bendy bonds, triple bonds use 3.

Molecule Builder

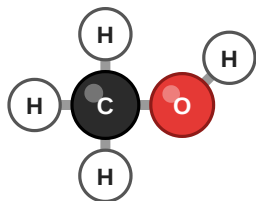
Level 2: Challenge Molecules

Bigger molecules with chains, rings and double bonds.
Can you build the Boss?

○ Hydrogen (H) ● Carbon (C) ● Oxygen (O) ● Nitrogen (N) ● Chlorine (Cl) — single bond — double/triple = bendy bonds

9 Methanol CH₃OH

☐ Built it!

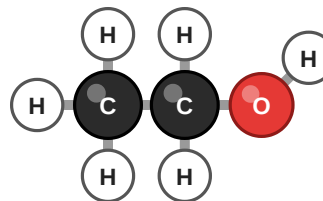


● 1 Carbon ● 1 Oxygen ○ 4 Hydrogen — 5 straight

Race-car fuel. Poisonous, so never drink it!

10 Ethanol C₂H₅OH

☐ Built it!

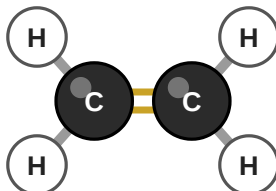


● 2 Carbon ● 1 Oxygen ○ 6 Hydrogen — 8 straight

The main ingredient in hand sanitizer.

11 Ethylene C₂H₄

☐ Built it!



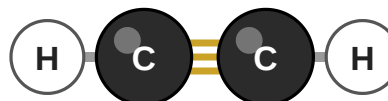
Shape: Flat

● 2 Carbon ○ 4 Hydrogen — 4 straight — 2 bendy

Bananas give off this gas to ripen. It is flat, not a pyramid!

12 Acetylene C₂H₂

☐ Built it!



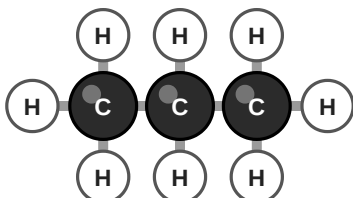
Shape: Straight line

● 2 Carbon ○ 2 Hydrogen — 2 straight — 3 bendy

Burns super hot in welding torches. Triple bond!

13 Propane C₃H₈

☐ Built it!



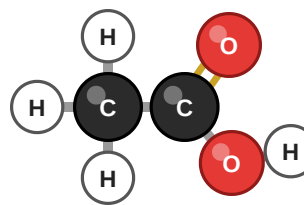
Shape: Zig-zag chain

● 3 Carbon ○ 8 Hydrogen — 10 straight

The fuel in a BBQ grill tank.

14 Vinegar (acetic acid) CH₃COOH

☐ Built it!

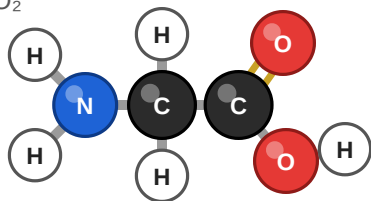


● 2 Carbon ● 2 Oxygen ○ 4 Hydrogen — 6 straight — 2 bendy

Makes vinegar sour. Mix with baking soda for a volcano!

15 Glycine C₂H₅NO₂

☐ Built it!

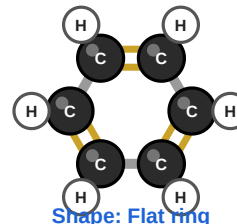


● 2 Carbon ● 2 Oxygen ● 1 Nitrogen ○ 5 Hydrogen — 8 straight — 2 bendy

The simplest amino acid: a building block of your muscles. Found on comets too!

16 Benzene C₆H₆

☐ Built it!



Shape: Flat ring

● 6 Carbon ○ 6 Hydrogen — 9 straight — 6 bendy

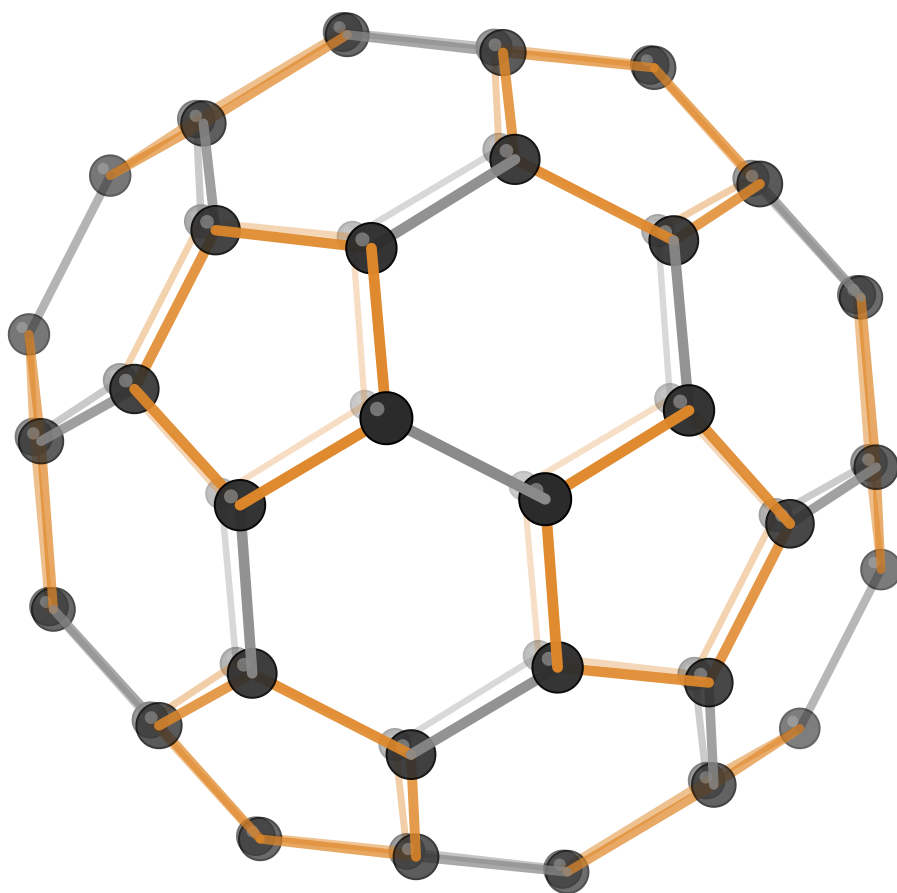
Boss level! A flat ring of 6 carbons. A chemist said he dreamed of a snake biting its tail.

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Bonus: The Buckyball (C_{60})

Your LINKTOR kit has a separate bag just for this: 60 carbons and 90 bonds.



60 carbon atoms

90 bonds

12 pentagons —

20 hexagons —

Same pattern as a soccer ball! Found in candle soot and even floating in outer space. Its discovery won the 1996 Nobel Prize in Chemistry.

- 1 Make a pentagon.** Connect 5 carbons in a ring (5 bonds). Every carbon in a buckyball joins exactly 3 others. ☐
- 2 Add 5 hexagons around it.** Each side of the pentagon becomes one side of a 6-carbon ring. It will start to curve into a bowl. ☐
- 3 Golden rule:** a pentagon never touches another pentagon. Each pentagon is surrounded by 5 hexagons. ☐
- 4 Keep going** around the bowl, adding pentagons and hexagons until it closes into a ball. Last check: 12 pentagons, 20 hexagons. ☐

Tip: work in pairs. One person holds the bowl while the other adds rings. If the ball won't close, look for two pentagons touching.